Grade 7 Name_Answer Key_________
Unit 9 Date______________________
Lesson 6 Practice Period____________________

1. Rewrite the expression $5 + (8 + 4)$ using the commutative property.

   $5 + (4 + 8)$

2. Rewrite the expression $5 + (8 + 4)$ using the associative property.

   $(5 + 8) + 4$

3. Rewrite the expression $5 + (8 + 4)$ using the distributive property.

   $5 + 8 + 4$

Problems 4-6 are equivalent to the expression $2 + (8 + 6) - (4 + 5) + 1$.
Without evaluating, determine which property (commutative, associative, or distributive) was used to re-write the expression $2 + (8 + 6) - (4 + 5) + 1$.

4. $(2 + 8) + 6 - (4 + 5) + 1$

   associative property

5. $2 + (6 + 8) - (4 + 5) + 1$

   commutative property

6. $2 + 8 + 6 - 4 - 5 + 1$

   distributive property

Determine whether the expressions are equivalent. If they are not, rewrite the second expression to make the expressions equivalent.

7. $7-(2.9+8)$

$7-10.9$

$=\ -3.9$

$7\ -2.9\ +(-8)$

$4.1+(-8)$

$=-3.9$

equivalent

8. $\left(2.6+\frac{4}{9}\right)+\ 5$

$\left(2\frac{6}{10}+\frac{4}{9}\right)+5\rightarrow\left(2\frac{54}{90}+\frac{40}{90}\right)+5$

$\rightarrow 2\frac{94}{90}+5\rightarrow 3\frac{4}{90}+5=8\frac{2}{45}$

$5+\left(2.6+\frac{4}{9}\right)$

$5+\left(2\frac{6}{10}+\frac{4}{9}\right)\rightarrow 5+\left(2\frac{54}{90}+\frac{40}{90}\right)$

$\rightarrow 5+2\frac{94}{90}\rightarrow 5+3\frac{4}{90}=8\frac{2}{45}$

associative property

equivalent

9. $\left(\frac{1}{8}\times 3\frac{1}{5}\right)\times\frac{1}{5}$

$\left(\frac{1}{8}\times\frac{16}{5}\right)\rightarrow\ \frac{16}{40}=\frac{2}{5}$

$\rightarrow\frac{2}{5}\times\frac{1}{5}=\frac{2}{25}$

$\frac{1}{8}\times\left(3\frac{1}{5}\times\frac{1}{5}\right)$

$\frac{1}{8}\times\left(\frac{16}{5}\times\frac{1}{5}\right)$

$\frac{1}{8}\times\frac{16}{25}=\frac{16}{200}=\frac{2}{25}$

equivalent

10. $(15\ -\ 3)\ -(6\ +8)$

$12-14$

-2

$(15\ -\ 6)\ +3\ -8$

$9+3-8\rightarrow 12-8$

-4

not equivalent

second expression should be rewritten as:

$(15-3)-(8+6)$